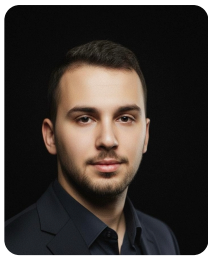
Muhammet Utku Küçük

Backend Engineer

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SUMMARY

Backend engineer with a B.Sc. in Computer Engineering (2025) who builds REST APIs and backend services. Built PartRadar, an event-driven microservices platform in **Java and Spring Boot** with Kafka, RabbitMQ and Keycloak, deployed to Kubernetes with Helm; before that shipped Node.js/TypeScript services on a remote contract at Covalent and PHP (Laravel) backends for corporate client projects. Comfortable with PostgreSQL, MySQL, Redis, Docker, Linux, Nginx and GitHub Actions CI/CD.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, PHP, C#, SQL, C/C++

Backend: Spring Boot, Spring Data JPA, Hibernate, Spring Security, Spring Cloud Gateway, Microservices, Apache Kafka, RabbitMQ, Keycloak, Node.js, Express.js, Laravel, Django, ASP.NET, REST APIs, OpenAPI/Swagger, JWT, RBAC, LDAP/Active Directory

Databases: PostgreSQL, PostGIS, MySQL, MSSQL, MongoDB, SQLite, Redis

Data Analysis: SQL (joins, aggregations, window functions), Python (Pandas, NumPy), Matplotlib, Jupyter Notebook, Excel (pivot tables), Power BI, ETL, data visualization

DevOps & Cloud: Kubernetes, Docker, Docker Compose, Helm, Linux, Nginx (reverse proxy, SSL/TLS), GitHub Actions (CI/CD), AWS (EC2, S3), Cloudflare

Testing & Tools: Unit Testing (JUnit, Jest, PHPUnit), Testcontainers, Postman, Git, Maven, Jira, Agile/Scrum

Frontend: React, Astro, Tailwind CSS, Bootstrap, jQuery

IoT: ESP32, ESPHome, MQTT, Home Assistant

EXPERIENCE

R&D Project Evaluator | Zaim Teknopark · Freelance · Remote — **Oct 2026 – Present**

- Evaluate R&D proposals and progress reports of software, cloud and IoT projects for technical feasibility, architecture, innovation and R&D eligibility, and recommend approval or revision.

Backend Developer | Covalent · Contract · Remote — **Sep 2025 – Feb 2026**

- Built and maintained Node.js/TypeScript backend services and REST endpoints for the Covalent Unified API, serving on-chain data (token balances, transactions, NFT metadata) to client applications.
- Developed backend features for a token/NFT marketplace dApp: listing and transaction-history endpoints, wallet-based sign-in, input validation and PostgreSQL data models.
- Added Redis response caching, pagination and rate-limit handling with retries for upstream calls; documented endpoints with OpenAPI, tested them with Jest and shipped through PR-based code reviews.

Software Developer (Computer Vision) | Nova UAV Team · University R&D Team · Istanbul — **Nov 2024 – Apr 2025**

- Worked on Python/OpenCV image-processing modules for UAV camera feeds (segmentation, object detection).

Software Developer (Backend) | Teknolojik Çözüm · Full-time, then part-time · Istanbul — **Aug 2022 – Apr 2023**

- Developed backend features, REST APIs and MySQL-backed admin panels with PHP (Laravel) and JavaScript (jQuery) for corporate client projects, including schema design and SQL reports.
- Containerized applications with Docker Compose and configured Nginx reverse proxies with SSL/TLS.
- Set up GitHub Actions CI/CD pipelines and solved production issues through log and root-cause analysis.

Software Engineering Intern | BİL Holding · Istanbul — **Jul 2021 – Aug 2021**

- Developed REST API endpoints and business logic in PHP (Laravel), plus responsive pages and forms (jQuery, Bootstrap) connected to them, for internal web applications.
- Designed MySQL tables, reporting screens and Excel exports; wrote PHPUnit tests and joined Git code reviews.

Software Development Intern | Kale Kalıp · Istanbul — Jul 2019 – Aug 2019

- Added modules and features to internal business applications in C# (.NET) backed by MSSQL.
- Wrote Python ETL scripts (Pandas, Matplotlib) for production-floor IoT sensor data, plus SQL and Excel pivot reports.

PROJECTS

PartRadar: Local Auto-Parts Finder — GitHub

Java 25, Spring Boot 4, Spring Cloud Gateway, Spring Security (Keycloak), Spring Data JPA, Apache Kafka, RabbitMQ, PostgreSQL + PostGIS, Testcontainers, Kubernetes, Helm, GitHub Actions

- Event-driven microservices platform that routes a driver's part request to nearby shops: Spring Cloud Gateway plus shop, inquiry and notification services with a database per service, Kafka domain events and RabbitMQ notifications with retries and a dead-letter queue.
- PostGIS nearby-shop search, a shared library for Keycloak JWT role mapping and error handling, Testcontainers integration tests and Helm deploys to Kubernetes with an end-to-end smoke test in CI.

Commerce Platform: E-Commerce Backend — In progress

Java 21, Spring Boot 3, Spring Data JPA, Hibernate, H2, Maven, Spring Boot Actuator

- E-commerce backend in progress: JPA domain model (users, addresses, roles, products, orders), DTOs, repositories and services.

Viducate: Video Learning Platform — GitHub · Live demo

Node.js, Express.js, PostgreSQL, Sequelize (ORM), FFmpeg, JWT, LDAP, Nginx, PM2, React

- Courses, quizzes and progress tracking for three roles: chunked uploads, byte-range streaming, FFmpeg transcoding.
- JWT auth with LDAP/Active Directory sign-in, role-based access, CSV bulk import and rate limiting; PM2 + Nginx.

IronEcho: Offline-First Fitness Platform — GitHub

Node.js, Express.js, MongoDB, Socket.io, JWT, Joi, React Native, TypeScript, Docker Compose

- Express REST API with JWT access/refresh tokens, Joi validation and real-time Socket.io updates; the React Native client queues offline writes and syncs them with last-write-wins conflict resolution.

HireHub: Role-Based Job Board — GitHub · Live demo

React 19, React Router 7, Tailwind CSS 4, Vite, GitHub Actions

- Job board for three roles with role-based route guards over a mocked REST API; deployed by GitHub Actions.

utkukucuk.com: Personal Portfolio — Website

Astro, TypeScript, Tailwind CSS, WebGL, GitHub Actions, Lighthouse CI, Cloudflare

- Bilingual (EN/TR) site with a WebGL hero and strict CSP; every PR passes lint, type-check and Lighthouse CI.

EDUCATION

B.Sc. in Computer Engineering (English) | Istanbul Aydın University — 2022 – 2025

Started at Bahçeşehir University, Computer Engineering (2020 – 2022); completed via lateral transfer.

Associate Degree in Computer Programming | Istanbul Aydın University — 2018 – 2020

Information Technologies (Software & Networking) | Avcılar Anatolian Technical High School — 2014 – 2018

CERTIFICATIONS

Claude Code Bootcamp: Hooks, MCP & Agentic AI Workflows | Udemy — 2026

Structuring Machine Learning Projects | DeepLearning.AI — 2020

LANGUAGES

Turkish (Native) · English (B2 – Professional Working Proficiency)

REFERENCES

Yasin Kara – Founder, Teknolojik Çözüm and Mahir Bilişim

Prof. Dr. Zafer Arslan – Department of Computer Engineering, Istanbul Aydın University

Serkan Seyhan – Co-Founder, Quantum Plasma Technology · Board Member, Bizim Bulut

Contact details available upon request.